Exact diffusion scores for Markov data with belief propagation

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Abstract

A diffusion model learns the score of noisy data. To understand what a network learns, one needs the exact score, which is rarely available. We study data generated by a stationary first-order Markov chain. Gaussian noise destroys the Markov property of the data at intermediate noise levels, but the posterior of the clean chain given the noisy one is still a chain. Belief propagation on this chain therefore gives the exact score for any transition kernel, also when no closed form exists, and we compute it on a grid. When the kernel is unknown, we learn it from noisy sequences alone by expectation–maximisation, with belief propagation as the E-step. On a linear chain with Laplace noise, this estimator trained on 32 sequences is more accurate than a convolutional network trained by denoising score matching on 2048. On snapshots of the stochastic Allen–Cahn equation, a kernel with five parameters reaches 3 to 17% of the error of the best linear denoiser, and its parameters give back the temperature and the double-well potential of the model.

1 Introduction

Diffusion models generate data by reversing a noising process, and the only object they learn is the score, the gradient of the log-density of the noisy data at each noise level (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021). A trained network does not reproduce its training set: it generalises (Kadkhodaie et al., 2024). To understand what it learns, it helps to know the exact score of the data distribution. The exact score explains the dynamical regimes of the backward process (Biroli et al., 2024) and when a model memorises (Bonnaire et al., 2025). These results use simple data models, each chosen to isolate one feature of real data: Gaussian mixtures for multimodality, trees for hierarchy (Sclocchi et al., 2025; Garnier-Brun et al., 2024).

Here the feature is order and locality. We take data generated by a stationary first-order Markov chain, a sequence in which each value depends only on the previous one. We ask what can be done with the exact score when the data are Markov. Three observations organise the paper, and Figure 1 shows one panel for each.

First, the noise breaks the Markov property. For a Gaussian chain the precision matrix of the noisy data is tridiagonal without noise, dense at intermediate noise, and close to diagonal at large noise (panel a). Second, the posterior of the clean chain given the noisy one is still a chain, so belief propagation (BP) computes the exact score. With the kernel learned from noisy data by expectation–maximisation (EM), this estimator needs far fewer samples than a network (panel b). Third, the method works on data from a physical model, snapshots of the stochastic Allen–Cahn equation, where the kernel has no closed form and five parameters suffice (panel c).

Our contributions are the following.

- For any first-order chain observed through Gaussian noise, the exact score at every noise level is given by BP on a grid, with an error that we measure and that is at round-off for smooth kernels (Section 3).

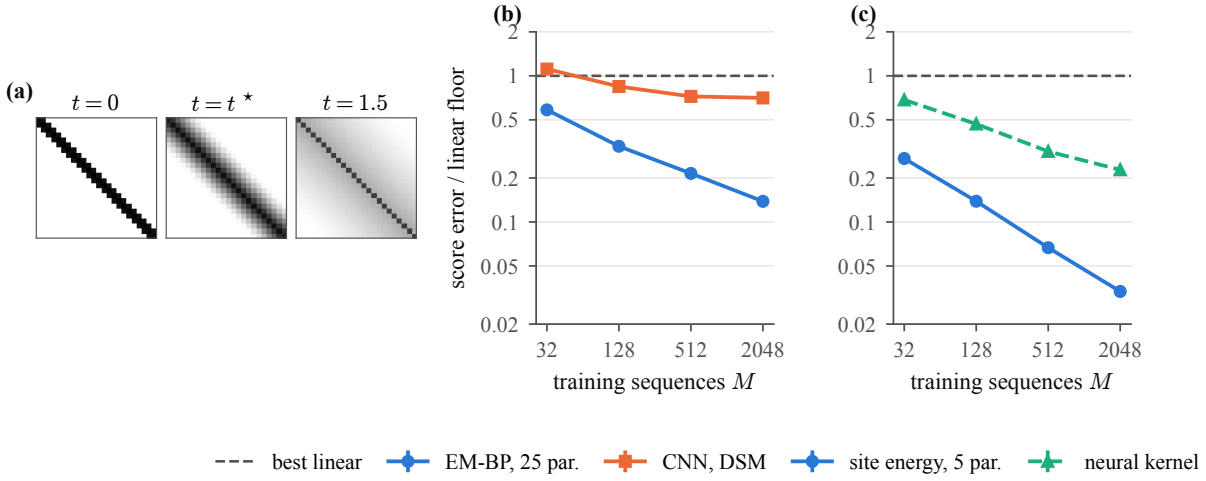


Figure 1: (a) Absolute values of the precision matrix Q_t of a noisy Gaussian chain ($\rho = 0.85$, $L = 24$, one logarithmic grey scale for the three panels) at $t = 0$, at $t^\star = 0.075$ where the entries beyond the first off-diagonal are largest, and at $t = 1.5$, where the off-diagonal entries are below 4% of the diagonal. (b) Laplace chain: score error of EM–BP and of a convolutional network trained by denoising score matching, divided by the error of the best linear denoiser, against the number M of training sequences. (c) The same for snapshots of the stochastic Allen–Cahn equation, with a five-parameter kernel and with a neural kernel inside BP. Mean over seeds, bars one standard error.

- The kernel can be learned from noisy sequences alone by EM through BP. With the right family of kernels this estimator beats a network trained on 64 times more sequences (Section 5).
- On the Allen–Cahn field the learned kernel is a physical model: its temperature and potential can be read off it, and two identities give the same readings from a neural kernel (Section 6).

2 Noise breaks the Markov property

The clean data are sequences $\mathbf{a} = (a_1, \dots, a_L)$ from a stationary chain,

$$p(\mathbf{a}) = \pi(a_1) \prod_{i=1}^{L-1} K(a_{i+1} \mid a_i),$$

with transition kernel K and stationary law π, standardised to unit variance. The noising process is the Ornstein–Uhlenbeck process, so at time t

$$\mathbf{x} = \alpha_t \mathbf{a} + \sqrt{\Delta_t}\, \mathbf{z}, \qquad \alpha_t = e^{-t}, \quad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(0, I).$$

The score $S_t(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is all a diffusion model needs, and by Tweedie's formula (Robbins, 1956; Efron, 2011) it is a function of the posterior mean,

$$S_t(\mathbf{x}) = -\frac{\mathbf{x} - \alpha_t\, \mathbb{E}[\mathbf{a} \mid \mathbf{x}]}{\Delta_t}. \tag{1}$$

The simplest chain is Gaussian, $a_{i+1} = \rho a_i + \sqrt{1-\rho^2}\, \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0,1)$. Its covariance is $\Sigma_0 = (\rho^{|i-j|})_{ij}$ and its precision $Q_0 = \Sigma_0^{-1}$ is tridiagonal: this is the Markov property written in algebra. The noisy data are Gaussian with covariance $\Sigma_t = \alpha_t^2 \Sigma_0 + \Delta_t I$ and their score is linear, $S_t(\mathbf{x}) = -Q_t \mathbf{x}$ with $Q_t = \Sigma_t^{-1}$. At $t = 0$, Q_t is tridiagonal; as $t \to \infty$ it tends to the identity; in between it is dense (Figure 1a). The noisy sequence is no longer a Markov chain, and the score at one site depends on the whole sequence. For a Gaussian chain this costs nothing, since the score is a matrix product. For any other chain the score is nonlinear and has no closed form. The next section shows how to compute it.

3 The exact score by belief propagation

Given $\mathbf{x}$, the posterior of the clean sequence is

$$p(\mathbf{a} \mid \mathbf{x}) \;\propto\; \pi(a_1) \prod_{i=1}^{L-1} K(a_{i+1} \mid a_i) \prod_{i=1}^{L} \ell_i(a_i), \qquad \ell_i(a) = \mathcal{N}(x_i; \alpha_t a, \Delta_t).$$

The noise acts on each site separately, so the posterior is again a chain. Its factor graph is a tree, and on a tree belief propagation, the sum–product algorithm (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009), is exact. It sends one message forward and one backward along the chain,

$$m_1^{\rightarrow}(a) = \pi(a), \qquad m_{i+1}^{\rightarrow}(a) = \int K(a \mid b)\, \ell_i(b)\, m_i^{\rightarrow}(b)\, db,$$

$$m_L^{\leftarrow}(a) = 1, \qquad m_{i-1}^{\leftarrow}(b) = \int K(a \mid b)\, \ell_i(a)\, m_i^{\leftarrow}(a)\, da,$$

and the marginal of site i is $p(a_i \mid \mathbf{x}) \propto m_i^{\rightarrow}(a_i)\, \ell_i(a_i)\, m_i^{\leftarrow}(a_i)$. Its mean, put into (1), gives the exact score at every site. The only input is the kernel K; no closed form of the score is needed. This is the same forward–backward recursion as for hidden Markov models (Rabiner, 1989), here with a continuous hidden state.

The grid. Each message is a function on $\mathbb{R}$. We represent it by its values on N_g equally spaced points of $[-A, A]$ and replace the integrals by the trapezoid rule, so that one update is a matrix–vector product with the matrix $K(u_j \mid u_k)$. A sequence costs $O(LN_g^2)$ operations. We use $N_g = 401$ and $A = 8$ (in units of the standard deviation) throughout, and we measure the error this causes. On a Gaussian chain the grid score matches the closed form to 2×10^{-13} from 81 points on; on the Allen–Cahn chain of Section 6 it matches a 1601-point reference to 2×10^{-11} at 101 points (Appendix A). On kernels with a kink, such as the Laplace noise of Section 5, the error decreases more slowly, as a power of N_g; it is 9×10^{-6} at 401 points, well below every learning error we report.

Why not Gaussian messages. A natural alternative to the grid is to keep each message Gaussian, as approximate message passing does in dense models (Thouless et al., 1977; Donoho et al., 2009). On a chain this gives nothing beyond a linear estimator.

Proposition 1. *Let the kernel be linear and Gaussian, $K(a' \mid a) = \mathcal{N}(a'; \rho a, q)$, and π Gaussian. Then every BP message is Gaussian, its variance does not depend on $\mathbf{x}$ and its mean is linear in $\mathbf{x}$. Hence $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = H_t \mathbf{x}$ for a fixed matrix H_t, and the score is linear.*

Proof. If $m_i^{\rightarrow} = \mathcal{N}(\mu_i, v_i)$, one forward step gives $m_{i+1}^{\rightarrow} = \mathcal{N}(\mu_{i+1}, v_{i+1})$ with

$$v_{i+1} = \rho^2 \big(v_i^{-1} + \alpha_t^2/\Delta_t\big)^{-1} + q, \qquad \mu_{i+1} = \rho \left(v_i^{-1} + \alpha_t^2/\Delta_t\right)^{-1} \big(\mu_i/v_i + \alpha_t x_i/\Delta_t\big).$$

The variances never involve $\mathbf{x}$, and the means are linear in $\mathbf{x}$ by induction. The backward messages are the same, and a marginal is the product of three Gaussians. □

Replacing a non-Gaussian chain by Gaussian messages therefore gives a linear estimator, and no linear estimator beats the linear minimum mean-square-error (LMMSE) one, $\mathbb{E}^{\text{lin}}[\mathbf{a} \mid \mathbf{x}] = \alpha_t \Sigma_0 \Sigma_t^{-1} \mathbf{x}$, whose score is $-\Sigma_t^{-1}\mathbf{x}$, the score of the Gaussian with the same covariance as the data. We call its error the *linear floor* and report every error relative to it. A ratio below one means that the estimator uses structure that no linear denoiser can use.

4 Learning the kernel through the noise

In practice K is unknown, and we have noisy sequences only. Each of the M training sequences is observed once, at one of twelve noise levels $t \in [0.05, 3]$ (log-spaced), the sequences being split evenly

between the levels. We fit a parametric kernel K_θ by maximising the likelihood of the noisy data, which BP computes as a by-product. The EM algorithm (Dempster et al., 1977) does this with BP as the E-step. From the current θ, BP gives the posterior of every pair (a_i, a_{i+1}); summed over sites and sequences, these give the expected pair counts $\Xi(a, a')$ on the grid. The M-step maximises $\sum_{a,a'} \Xi(a, a') \log K_\theta(a' \mid a)$. The gradient of the noisy-data log-likelihood is the posterior average of $\nabla_\theta \log K_\theta$ (Fisher's identity), so the procedure is maximum likelihood without ever seeing a clean sequence. The learned kernel is then used in BP and (1) to give the score.

We compare three estimators, each knowing less about the data than the previous one.

1. **EM–BP with a kernel family of the right shape.** For the linear chain of Section 5, $K_\theta(a' \mid a) = \sum_{c=1}^{8} w_c \mathcal{N}(a' - \rho a; \mu_c, s_c^2)$, a linear drift with a mixture of eight Gaussians as noise (25 parameters). For the Allen–Cahn field, the five-parameter kernel of Section 6.

2. **EM–BP with a neural kernel.** A mixture density network (Bishop, 1994),

$$K_\theta(a' \mid a) = \sum_{c=1}^{4} w_c(a)\, \mathcal{N}\big(a'; \mu_c(a), s_c(a)^2\big),$$

   whose weights, means and variances are outputs of a small network (two hidden layers of 32 units, 1,516 parameters). It assumes only that the data are Markov.

3. **A score network.** A convolutional network trained by denoising score matching (Vincent, 2011; Ho et al., 2020): at every step a clean training sequence, a noise level and the noise are drawn afresh. It assumes nothing about the data except locality, and it sees the clean sequences, which EM–BP never does.

Each estimator stops at the checkpoint with the lowest score error on a validation set. The error of an estimator is the relative score error $R = \big(\sum \|\hat{S}_t(\mathbf{x}) - S_t(\mathbf{x})\|^2 / \sum \|S_t(\mathbf{x})\|^2\big)^{1/2}$, summed over 2048 test sequences and the twelve levels, with the reference S_t from BP with the true kernel. Details are in Appendix C.

5 The Laplace chain

We take $a_{i+1} = \rho a_i + \varepsilon_i$ with Laplace noise ε_i of variance $1 - \rho^2$, $\rho = 0.85$ and $L = 32$. The chain is Markov and close to the Gaussian one, but its score is nonlinear and has no closed form. The linear floor is $R = 0.167$. Figure 2a shows the three estimators against the number of training sequences, over 16 seeds.

EM–BP is below the floor already at $M = 32$ (ratio 0.59) and reaches 0.14 at $M = 2048$. The network is above the floor at $M = 32$ (1.11) and levels off near 0.71. With 32 sequences EM–BP is more accurate than the network with 2048. The convolutional network is not the strongest architecture on this task: among six that we tried, the best reaches $R = 0.060$ at $M = 2048$, still above EM–BP at $M = 128$ ($R = 0.055$). The neural kernel lies between the two at large M (0.40 at 2048) but is above the floor at $M = 32$: a free conditional density needs data that a parsimonious family does not.

Panel b separates the noise levels. EM–BP is more accurate than the network at every level. The linear floor is better than the network above $t \approx 0.2$ and better than EM–BP above $t \approx 1$; there the noisy data are nearly Gaussian, the true score is nearly linear, and the small error of a learned kernel is what remains.

6 The stochastic Allen–Cahn field

The model. The stochastic Allen–Cahn equation (Allen and Cahn, 1979), $\partial_\tau \phi = \partial_x^2 \phi - V'(\phi) + \sqrt{2\Theta}\, \eta(x, \tau)$ with $V(\phi) = (\phi^2 - 1)^2/4$ and space–time white noise η, has the stationary law $\propto e^{-H[\phi]/\Theta}$

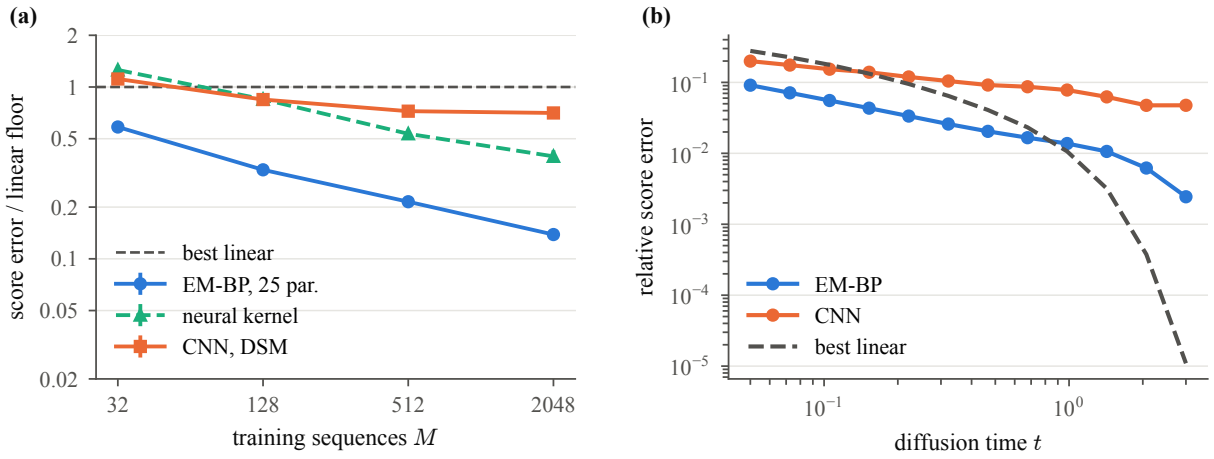


Figure 2: Laplace chain ($\rho = 0.85$, $L = 32$). (a) Score error divided by the linear floor against the number of training sequences, for EM–BP with the 25-parameter kernel, EM–BP with a neural kernel, and a convolutional network trained by denoising score matching. (b) Score error per noise level at $M = 128$.

with $H[\phi] = \int [\frac{1}{2}(\partial_x \phi)^2 + V(\phi)]\, dx$. On a lattice of spacing h the energy is $\sum_i [(\phi_{i+1} - \phi_i)^2/2h + hV(\phi_i)]$, a sum of nearest-neighbour terms, so a spatial snapshot is exactly a first-order Markov chain in the site index. Its kernel comes from the transfer operator

$$T(\phi, \phi') = \exp\Big(-\frac{h}{2\Theta}\big[V(\phi) + V(\phi')\big] - \frac{(\phi' - \phi)^2}{2\Theta h}\Big), \qquad K(\phi' \mid \phi) = \frac{T(\phi, \phi')\, \psi_0(\phi')}{\lambda_0\, \psi_0(\phi)},$$

with λ_0 the largest eigenvalue of T and ψ_0 its eigenfunction (Scalapino et al., 1972). The chain is reversible, the stationary law is ψ_0^2, the partition function of L sites satisfies $\log Z_L = L \log \lambda_0 + O(1)$, and the correlation length is $\xi = 1/\log(\lambda_0/\lambda_1)$ sites. For a quadratic V these quantities have a closed form; for the double well they do not, since the continuum limit is the Schrödinger problem of a quartic double well (Krumhansl and Schrieffer, 1975; Khare et al., 1997). We compute them by a dense eigensolve, which converges to 10^{-13} and reproduces the Gaussian closed form to 2×10^{-14} (Appendix B). Cold fields are made of domains near $\phi = \pm 1$ separated by kinks.

We use six cells: $\Theta \in \{0.3, 0.5, 1, 2\}$ with h chosen so that $\xi \approx 6$ sites ($L = 32$), and $h \in \{1, \frac{1}{2}, \frac{1}{4}\}$ at $\Theta = 0.3$ with the physical length fixed ($L = 32, 64, 128$).

A five-parameter kernel. The lattice energy is a quadratic bond plus a quartic site term, so the pair energy

$$W_\theta(a, a') = \frac{(a' - a)^2}{2\gamma} + f(a) + f(a'), \qquad f(a) = \sum_{m=1}^{4} c_m a^m,$$

with five parameters $(\gamma, c_1, \ldots, c_4)$, contains the true model exactly ($a = \phi/s$ is the standardised field). Its kernel is obtained from e^{-W_θ} as above, by one eigensolve on the grid. We call it the *site-energy kernel*. The mixture kernel of Section 5 is misspecified here: its drift is linear, and a linear drift cannot hold a domain.

Results. Table 1 and Figure 1c give the errors. At $M = 2048$ the site-energy kernel is at 3 to 17% of the linear floor on every cell, the neural kernel at 23 to 84%, and the mixture kernel at the floor (1.00 to 1.02). With $M = 32$ sequences the site-energy kernel is already at 24 to 28% of the floor on the three cells at $\Theta = 0.3$; on the two hottest cells, where the field is closer to Gaussian, it needs $M = 128$ to go below the floor.

Reading the physics off a kernel. Every kernel built from a pair energy of the form W above, whatever the site term f, satisfies two identities:

$$\frac{\partial^2 \log K(a' \mid a)}{\partial a\, \partial a'} = \frac{1}{\gamma}, \qquad f(a) = -\tfrac{1}{2} \log K(a \mid a) + \text{const}.$$

Table 1: PROVISIONAL (CNN pending). Score error divided by the linear floor, field cells.

	$M = 32$			$M = 2048$		
cell	site	neural	CNN	site	neural	CNN
$\Theta = 0.3$, $h = 1$	0.27	0.68	–	0.03	0.23	–
$\Theta = 0.3$, $h = \frac{1}{2}$	0.28	0.73	–	0.04	0.31	–
$\Theta = 0.3$, $h = \frac{1}{4}$	0.24	0.97	–	0.03	0.42	–
$\Theta = 0.5$	0.66	1.30	–	0.07	0.42	–
$\Theta = 1$	1.37	2.85	–	0.14	0.74	–
$\Theta = 2$	1.43	3.83	–	0.17	0.84	–

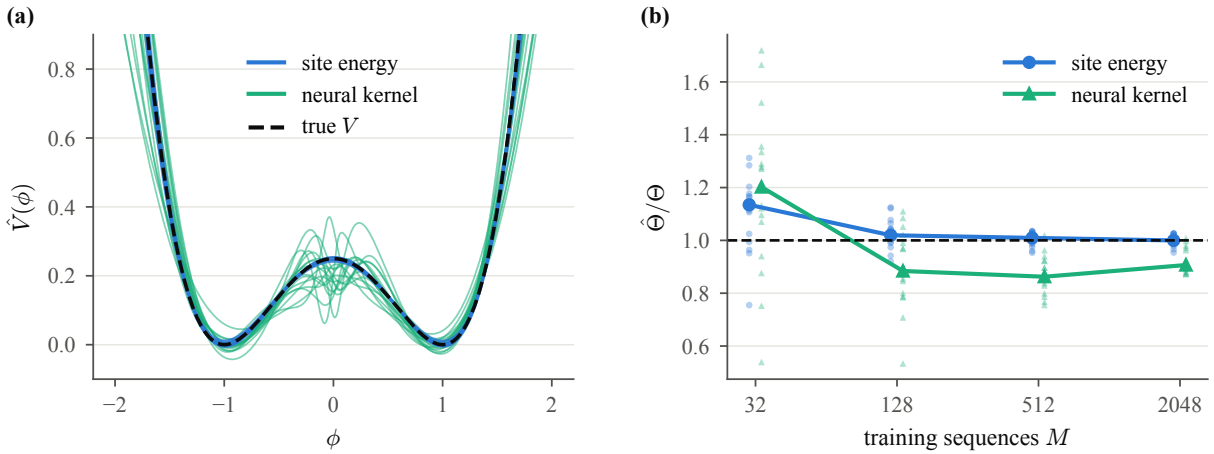


Figure 3: The physics read off kernels learned from noisy data, base cell. (a) The potential from the diagonal of the kernel, one curve per seed (16), $M = 2048$, for the site-energy kernel and the neural kernel; dashed, the true double well. (b) The temperature from the mixed derivative of log K; points are seeds, lines join the medians.

The first holds because log K is the bond plus a function of a plus a function of a'; the second because the bond vanishes on the diagonal. With $\gamma = \Theta h/s^2$ they give the temperature $\hat{\Theta} = \hat{\gamma} s^2/h$ and the potential $\hat{V}(\phi) = (2\hat{\Theta}/h)\hat{f}(\phi/s)$. For the site-energy kernel this is its own parametrisation: at $M = 2048$ on the base cell ($\Theta = 0.3$, $h = 1$) the median $\hat{\Theta}/\Theta$ over seeds is 1.000 and the barrier is 0.248 (true value 0.25).

The identities use no parameter of the kernel, so they apply to the neural kernel too, where the mixed derivative is no longer constant and we average it under the kernel's stationary pair law. The temperature comes back within 9% on all six cells at $M = 2048$ ($\hat{\Theta}/\Theta$ from 0.91 to 1.00). The potential comes back on the base cell (barrier 0.20; Figure 3a) but not on fine or hot lattices, where the factor $2\Theta/h$, up to 27, multiplies the error of the kernel's diagonal. A network that knows only that the data are Markov thus recovers part of the physics.

7 Discussion

When data are Markov, the exact score is within reach even though the noisy data are not Markov: BP on the posterior chain computes it for any kernel, and EM through BP learns the kernel from noisy data alone. The advantage over a network is data efficiency. A kernel with five or 25 parameters cannot memorise even one training sequence of 32 values, and it reaches errors that a network does not reach with 64 times more data. On the Allen–Cahn field the learned parameters are the temperature and the potential of the model.

The method has limits. The data are one-dimensional sequences with first-order dependence. The kernel family is a modelling choice: the mixture kernel that works on the linear chain fails on the Allen–Cahn field. The neural kernel removes this choice at the price of more data. The grid costs N_g^2 per site,

and a very irregular kernel would need more points. Checkpoints were selected with the exact score on a validation set; with the held-out likelihood instead, which needs no exact score, the site-energy kernel is at 4 to 23% of the floor instead of 3 to 17%.

Two directions follow. One is model inference: from a few noisy time series of interacting units, a Markov kernel with few parameters can be learned and read. The other is to ask whether networks trained on Markov data learn to implement BP, as they do on trees (Garnier-Brun et al., 2024; Mei, 2024).

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A Accuracy of the grid

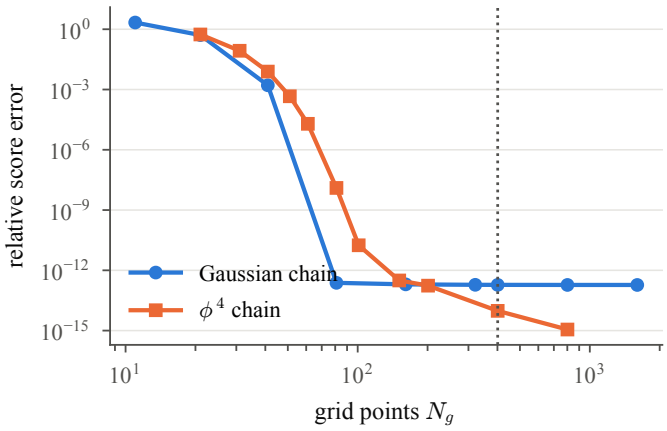


Figure 4: Relative score error of grid BP against the number of grid points, on $[-8, 8]$: the Gaussian chain ($\rho = 0.85$) against its closed-form score, and the Allen–Cahn base cell against a 1601-point reference; maximum over noise levels. Dotted: $N_g = 401$.

For smooth kernels the error falls faster than any power of N_g and reaches round-off by 81 points (Gaussian) and 151 points (Allen–Cahn). For the Laplace chain, whose kernel has a kink, the error against a 3201-point reference on a wider domain is 4×10^{-5} at 201 points and 9×10^{-6} at 401.

B $\log Z$ and the correlation length

For a chain of L sites, $\log Z_L = L \log \lambda_0 + O(1)$. Table 2 gives $\log \lambda_0$ and ξ at $h = 1$. They change by less than 2×10^{-13} from 2401 to 4801 quadrature points on $[-12, 12]$; with V replaced by a quadratic potential the same solver matches the closed form $\lambda_0 = \sqrt{2\pi/(a+c)}$, $a = hk/2\Theta + 1/\Theta h$, $c = \sqrt{a^2 - (\Theta h)^{-2}}$, to 2×10^{-14}. As Θ falls, $\log \lambda_0$ approaches the value for a single harmonic well and on fine lattices ξ grows like $e^{E/\Theta}$ with a fitted $E = 0.99$, close to the energy of one kink in the continuum, $2\sqrt{2}/3 \approx 0.94$.

Table 2: Free energy per site and correlation length of the lattice field, $h = 1$.

Θ	0.2	0.3	0.5	0.7	1	1.5	2
$\log \lambda_0$	-0.467	-0.186	0.161	0.362	0.550	0.737	0.857
ξ (sites)	19.47	6.08	2.75	1.98	1.53	1.21	1.06

C Protocol

Data. For each seed: M training sequences, 1024 validation and 2048 test sequences, each test sequence noised at all twelve levels. Laplace chain: 16 seeds. Allen–Cahn field: 16 seeds for the kernels, 8 for the network.

EM–BP. Up to 1200 EM iterations, checkpoints along the way, the one with the lowest validation error kept. The mixture kernel starts from $\rho = 0.3$; the neural kernel's M-step is one Adam step with backtracking on the M-step objective, so each EM iteration still increases the likelihood; the site-energy kernel's M-step is a quasi-Newton step with the gradient of the eigensolve from first-order perturbation theory.

Network. One convolution of width $2r + 1$ with 64 channels, a sinusoidal embedding of t added to it, then two pointwise layers with tanh, predicting the noise $\mathbf{z}$ at every site. Radius $r = 4$ on the Laplace chain (5,505 parameters). On the field $r = 2, 4, 8$ at $h = 1, \frac{1}{2}, \frac{1}{4}$ (the same physical reach) and $r = 4$ on the three hotter cells (128 channels at $\Theta = 0.5$), chosen on two development seeds. Adam with learning rate 10^{-3}, batch 64, gradient clipping at 1; 32,000 steps on the Laplace chain and 64,000 on the field, checkpoints at powers of two from 500 steps.

D Speciation, site by site

As t decreases during generation, the posterior of a cold field commits each site to one well. We followed exact Ornstein–Uhlenbeck paths of clean chains ($L = 256$, read backwards they have the law of a generation with the exact score) and recorded, for each site, the noise level below which BP gives its correct sign with probability at least 0.75 (Figure 5). The bulk of a long domain commits at once. The kinks are located last, by a front that moves from the bulk to the wall over a few sites: at $\Theta = 0.2$ the commitment time is 1.46 ten sites inside a domain and 0.44 at the kink. Short domains commit last, and nothing propagates across kinks.

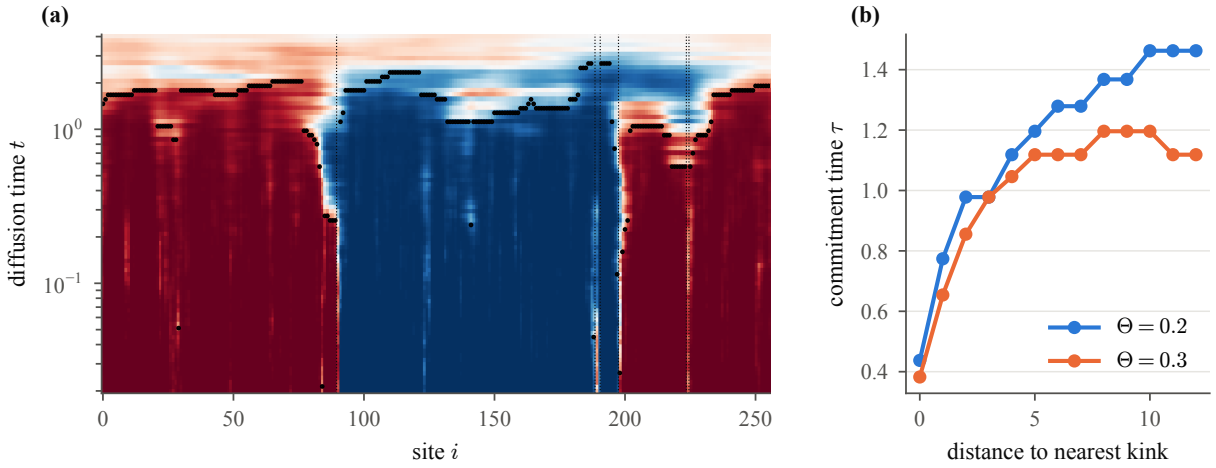


Figure 5: (a) Polarisation $2P(a_i > 0 \mid \mathbf{x}_t) - 1$ of one chain at $\Theta = 0.2$ ($\xi = 19.5$ sites), red positive and blue negative; dots, commitment times; dotted lines, kinks of the clean chain. (b) Median commitment time against the distance to the nearest kink, domains of at least 8 sites, 32 chains per temperature.